

ASSIGNMENTS**Class-5**

1. If w_1, w_2, w_3 are independent vectors, show that the sums $v_1 = w_2 + w_3, v_2 = w_1 + w_3, v_3 = w_1 + w_2$ are independent.
2. If v_1, v_2, v_3 are linearly independent, determine whether the vectors $v_1 - v_2, v_2 - v_3, v_3 - v_1$ are linearly independent or dependent.
3. Find the largest possible number of independent vectors among

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, v_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, v_5 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

4. True or False:

“If the columns of a matrix are linearly independent, then $AX=b$ has exactly one solution for every b ”